

Northern University of Business and Technology Khulna
Department of Computer Science and Engineering

WEEK 12: USER MANUAL

Project Title: AI-Powered Resume Analyzer and Career Copilot (RESUMIND)

Course Title: Mobile Computing Lab

Course Code: CSE4204

Submitted to:
Md. Riaz Mahmud
Assistant Professor
Department of Computer Science and Engineering
Northern University of Business and Technology, Khulna

Submission Date: 22 October 2026

TEAM INFORMATION SHEET

Field	Details
Official Team Name	CSE4204-8A-T05
Section	8A
Project Title	AI-Powered Resume Analyzer and Career Copilot (RESUMIND)
Team Leader	Mahammad Hasan (ID: 11220320828)
Team Member List with IDs	1. Mahammad Hasan (ID: 11220320828) - Team Leader / Frontend Developer + UI/UX Designer 2. Shanawaz Sakib (ID: 11220320916) - Core System Integration Engineer 3. Ashikur Rahman Himel (ID: 11220320833) - Database Storage & Quality Controller 4. Rasel Ratul (ID: 11220320827) - Backend Developer
Deployment Link (Live URL)	https://ai-resume-analyzer-psi-lime.vercel.app
GitHub Repository Link	https://github.com/mahamud894/AI_Resume_Analyzer
Figma Link	https://www.figma.com/design/Ls0B3DIjxRKrP2RIzYQxPG/AI-Resume-Analyzer?node-id=2-2&t=lFEVQCoUHt0JSumI-1

1. Introduction

This manual explains how to install, open and use **RESUMIND**, the AI Resume Analyzer built by team CSE4204-8A-T05. RESUMIND checks your resume against a job, tells you what to improve, rewrites it as a European-standard (Europass) CV and writes a matching cover letter.

Table 1: What you can do with RESUMIND

Feature	What it gives you
Resume analysis	Overall score (0–100) and five category scores: ATS, Tone & Style, Content, Structure, Skills — each with tips
Europass CV	Your resume rewritten in the European CV standard, with a list of every change and the reason
Before / after score	The improved CV scored for the same job, so you can see the improvement
Cover letter	A one-page letter for the same job, in formal or modern tone, that you can edit
History	All your resumes and analyses in one dashboard
Downloads	Report PDF, Europass CV PDF and cover letter PDF

1.1 Two ways to use it

- **Online (recommended):** open https://ai-resume-analyzer-psi-lime.vercel.app in any modern browser. Nothing to install.
- **Locally:** run the frontend and backend on your own computer (Chapter 2). This is for developers and examiners who want to see the code running.

1.2 Requirements

	Minimum
Device	Phone, tablet or computer with a screen width of 360 px or more
Browser	Recent Chrome, Edge, Firefox or Safari with JavaScript enabled
Internet	Required (the AI runs on Google's servers)
Resume	A text-based PDF, up to 10 MB (scanned images cannot be read)

2. Installation (Local Setup)

Skip this chapter if you use the online version.

2.1 Prerequisites

- **Node.js 22 or newer** and npm — nodejs.org
- **Git** — to clone the repository
- **A MongoDB Atlas cluster** (free M0) or a local MongoDB server
- **A Google Gemini API key** — free from Google AI Studio
- Optional: a Google OAuth client id (for "Sign in with Google") and Docker Desktop

2.2 Get the code

```
git clone https://github.com/mahamud894/AI_Resume_Analyzer.git
cd AI_Resume_Analyzer
```

2.3 Start the backend

```
cd backend
npm install
cp .env.example .env          # Windows PowerShell: Copy-Item .env.example .env
# open .env and fill in the values below
npm run seed                  # optional: demo user + sample resume
npm run dev                   # API on http://localhost:5000
```

Table 2: Backend environment variables (backend/.env)

Variable	Required	Example / description
`MONGO_URI`	Yes	mongodb+srv:// user:pass@cluster0.xxxx.mongodb.net/resumind
`JWT_SECRET`	Yes	A long random string (e.g. 96 hex characters)
`GEMINI_API_KEY`	For AI	AIza… key from Google AI Studio
`CLIENT_URL`	No	http://localhost:5173 (frontend address allowed by CORS)
`GOOGLE_CLIENT_ID`	No	OAuth client id for Sign in with Google
`PORT` · `JWT_EXPIRES_IN` · `GEMINI_MODEL` · `MAX_FILE_SIZE_MB`	No	5000 · 7d · gemini-flash-latest · 10

Tip: generate a JWT secret with the command below. Never commit the `.env` file — it is already in `.gitignore`.

```
node -e "console.log(require('crypto').randomBytes(48).toString('hex'))"
```

2.4 Start the frontend (second terminal)

```
cd frontend
npm install
cp .env.example .env          # VITE_API_URL=http://localhost:5000
npm run dev                   # open http://localhost:5173
```

2.5 Alternative: Docker Compose

```
# from the project root, after filling in backend/.env
docker compose --env-file backend/.env up --build
# frontend → http://localhost:3000, backend → http://localhost:5000
```

2.6 Check that it works

1. Open http://localhost:5000/api/health — you should see `"status":"ok"`.
2. Open http://localhost:5173 — the RESUMIND landing page appears.
3. Optional: run the automated tests with `cd backend && npm test` (48 tests).

3. Login Process

3.1 Create an account

1. On the landing page click **Get Started**.
2. Switch to **Sign Up**.
3. Enter a username (3–30 letters, numbers, _ or .), your e-mail and a password of **at least 8 characters**.
4. Click **Sign Up**. You are logged in and taken to your dashboard.

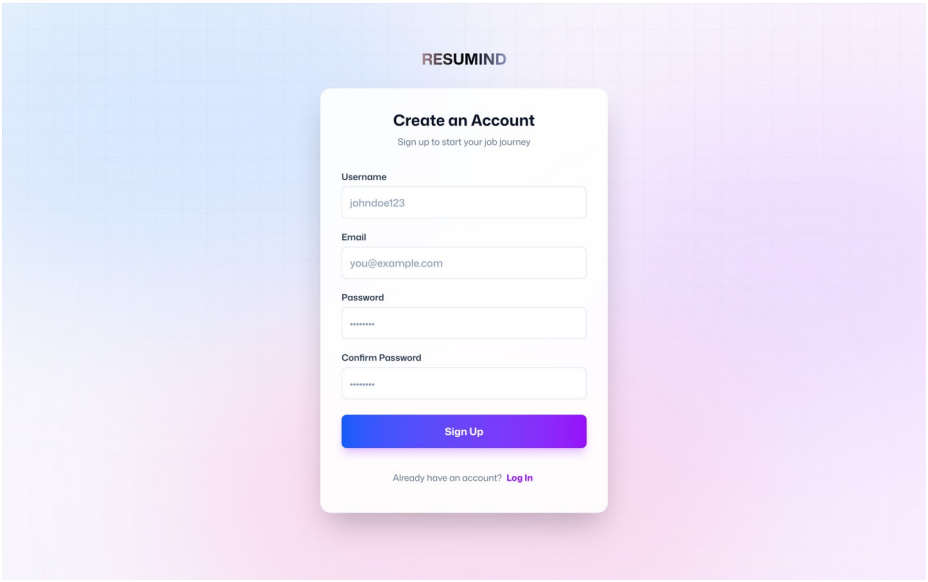


Figure 1: Sign-up dialog

3.2 Log in

1. Open the login dialog, enter your e-mail and password and press **Log In**.
2. Or click **Sign in with Google** and choose your Google account — no password needed.
3. You stay logged in on this device for 7 days.

Security: After 20 failed attempts in 15 minutes, login is paused for a while to protect your account.

3.3 Log out

Click **Log Out** in the top bar (on a phone, open the ☰ menu first). You return to the landing page.

3.4 Manage your account

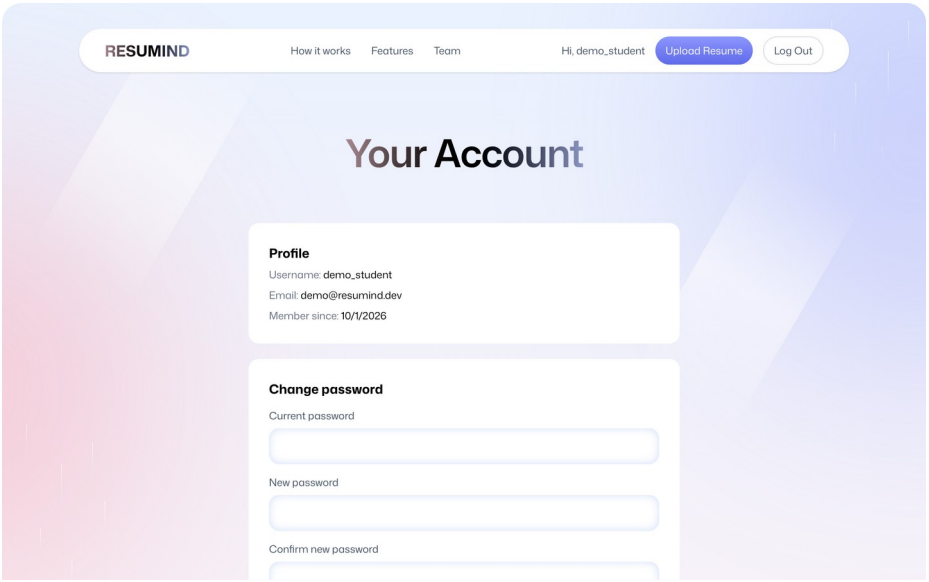


Figure 2: Account page

- **Change password:** Click **Hi, <username>** in the top bar to open the Account page → **Change password** → enter the current and the new password (min. 8 characters). All other devices are logged out automatically.
- **Google-only accounts** can set a first password here, so they can also log in with e-mail.
- **Delete account:** Account → Delete account → confirm with your password (or your e-mail for Google-only accounts). All resumes, analyses, CVs, letters and files are permanently removed.

4. Features and System Usage

The usual workflow has four steps, shown by the progress bar at the top of each page:

Step 1	Step 2	Step 3	Step 4
Upload resume + job	Read the analysis	Improve to Europass CV	Generate cover letter

4.1 Dashboard

After login the dashboard lists all your resumes, newest first. Each card shows the company, the job title and the score. Click a card to open its result, or click **Upload Resume** to start a new analysis. Use the delete icon on a card to remove a resume and everything created from it.

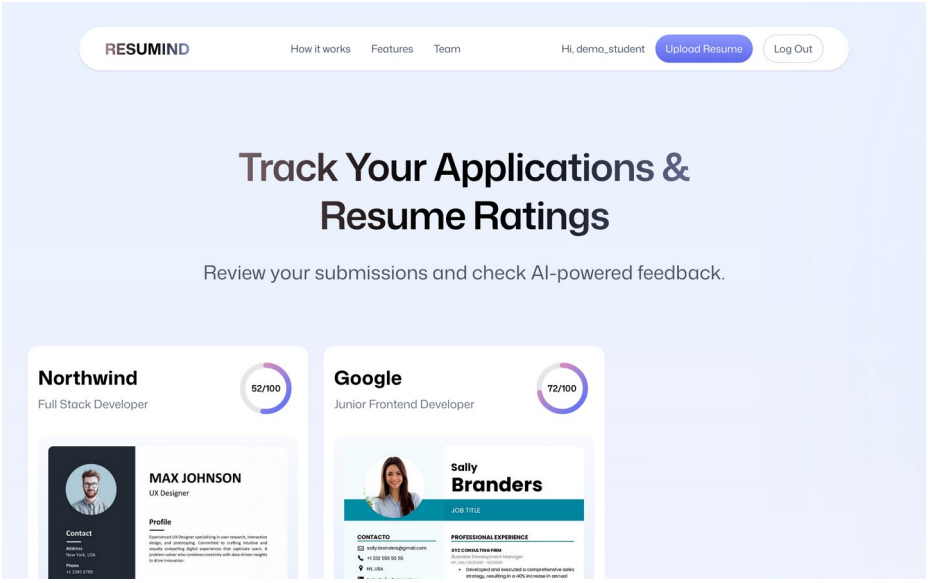


Figure 3: Dashboard

4.2 Upload and analyse a resume

1. Click **Upload Resume** in the top bar.
2. Enter the **company name** (optional), the **job title** and paste the **job description** (at least 20 characters).
3. Drag your PDF onto the upload box or click it to choose the file.
4. Click **Analyze Resume**. The page shows the progress: uploading → reading → analysing. This usually takes 10–20 seconds.
5. When it is finished you are taken to the result page automatically.

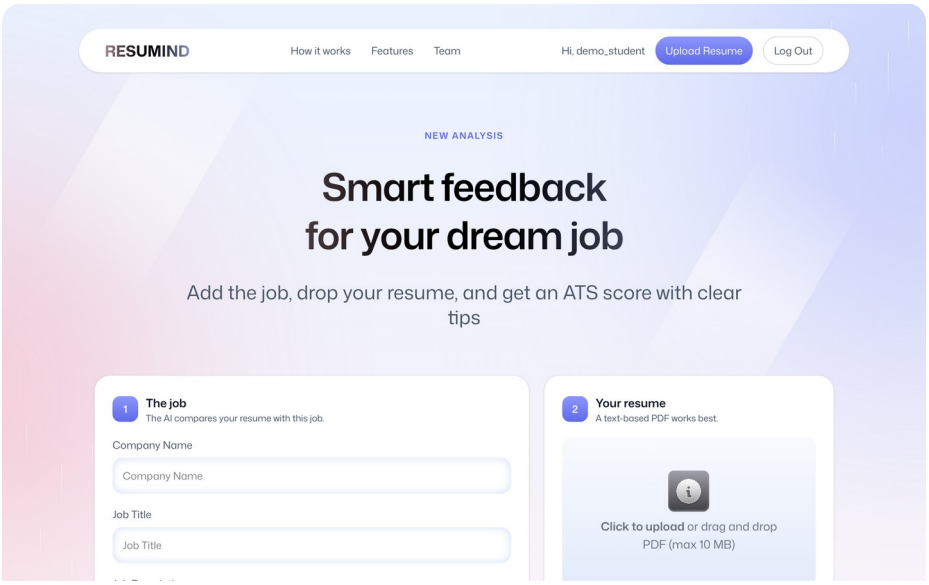


Figure 4: Upload page

4.3 Read your results

- The **overall score** (0–100) is shown in a gauge; green is good, amber is average, red needs work.
- The **ATS** card shows how well an Applicant Tracking System can read your resume.
- Open each category (Tone & Style, Content, Structure, Skills) to see ✓ good points and ! things to improve with explanations.
- On the left you see a preview of your resume.
- Click **Download Report** to save a printable PDF of the analysis.
- Use **Analyze this resume for another job** to check the same resume against a different job.

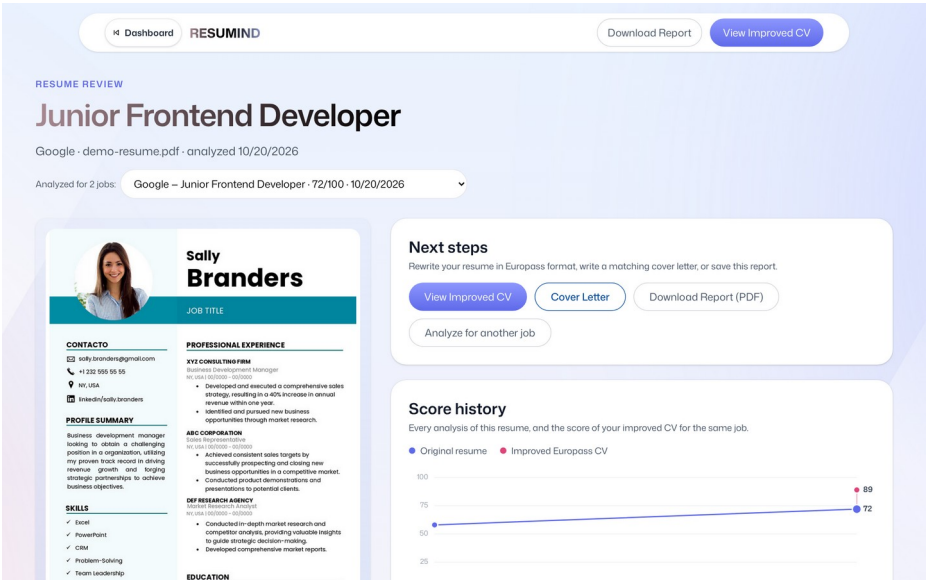


Figure 5: Result page

4.4 Improve to a Europass CV

1. On the result page click **Improve My CV (Europass)** (later: **View Improved CV**). This takes about 20–40 seconds.
2. Read the CV. Text in square brackets, e.g. `[Add your phone number]`, is information that was missing — the AI never invents facts.
3. Click any text in the CV to **edit** it directly. Your changes are saved; **Reset to AI version** restores the original.
4. Scroll to **What we improved** to see every change and why it was made.
5. Click **Download PDF** to save the CV as an A4 PDF.

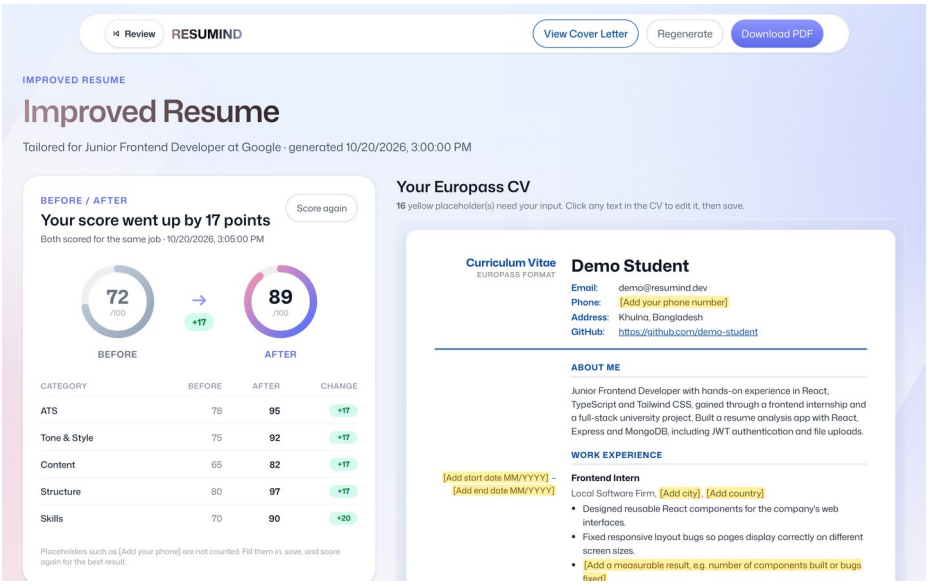


Figure 6: Europass CV page

4.5 Before / after score

On the Europass page click **Score my improved CV**. The improved CV (including your edits) is scored for the same job. The result page then shows the old and new score for each category and a score-history chart.

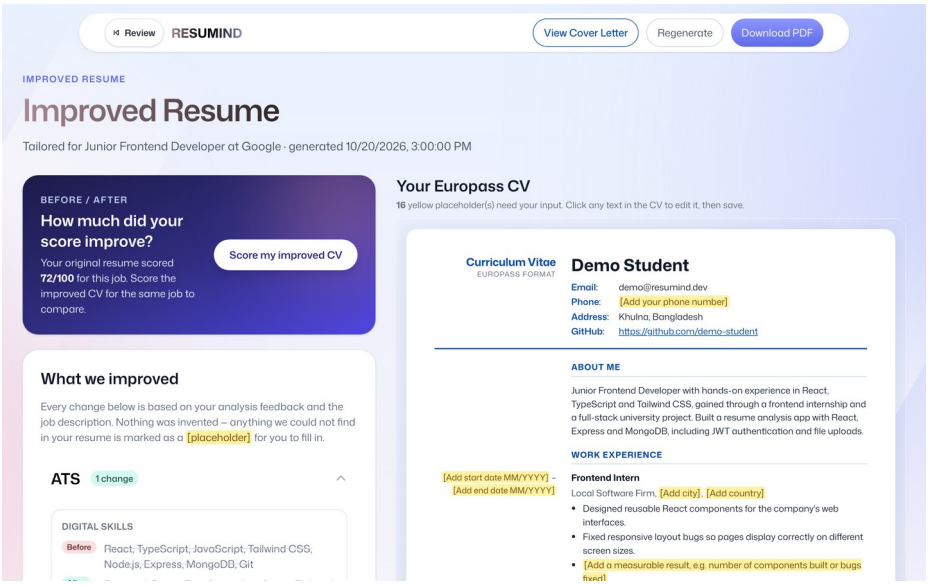


Figure 7: Before / after comparison

4.6 Generate a cover letter

1. On the Europass page click **Generate Cover Letter**.
2. Optionally enter the hiring manager's name (e.g. Ms. Sarah Rahman), the company address and where you found the job (e.g. LinkedIn).
3. Choose the tone: **Formal** ("Yours sincerely") or **Modern** ("Kind regards").
4. Click **Generate Cover Letter**. The letter appears as an A4 page; click it to edit the text.
5. Replace any `[placeholder]`, then click **Download PDF**.

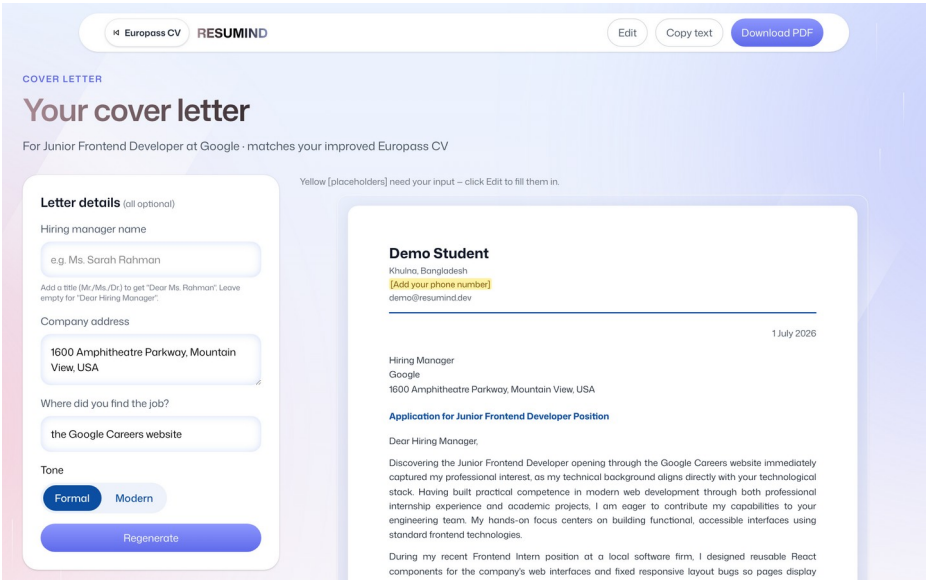


Figure 8: Cover letter page

4.7 Using RESUMIND on a phone

Every page adapts to small screens: the menu collapses, cards stack in one column and the resume preview moves below the feedback. You can upload a PDF directly from your phone's files or Google Drive.

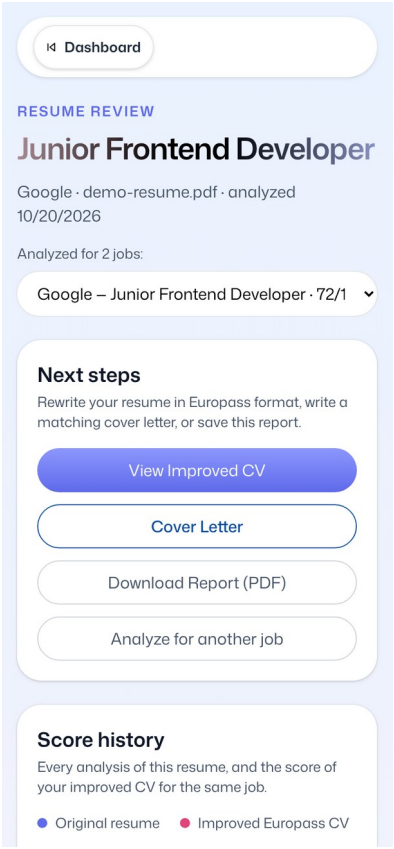


Figure 9: Result page on a phone

5. Troubleshooting

Table 3: Common messages and what to do

Message / problem	Reason	What to do
First load takes up to a minute	The free server sleeps after 15 minutes	Wait once; afterwards it is fast
"This PDF has no readable text"	The PDF is a scanned image	Export the resume from Word/Google Docs as PDF
"Only PDF files are allowed"	Wrong file type or renamed file	Upload a real .pdf file
"File too large"	File is over 10 MB	Compress the PDF or remove images
"The AI model is busy"	Gemini is overloaded	Try again after a minute
"Too many requests"	15 AI requests per hour reached	Wait until the hour is over

Message / problem	Reason	What to do
Logged out unexpectedly	Password was changed or token expired (7 days)	Log in again
Google button missing (local)	VITE_GOOGLE_CLIENT_ID not set	Add it to frontend/.env and restart

6. Frequently Asked Questions

Is my data private?

Yes. Every resume, analysis, CV and letter is linked to your account and can only be read with your login token. Passwords are stored only as bcrypt hashes. You can delete everything at any time.

Will the AI add experience I don't have?

No. The prompt forbids inventing facts and the backend replaces any new number or date with a placeholder for you to fill in.

Why does my score change slightly if I analyse again?

The scores come from an AI model, so small differences (a few points) are normal. Large improvements come from following the tips.

What is Europass?

Europass is the European Union's standard CV format, widely used for jobs, scholarships and higher studies in Europe. It uses MM/YYYY dates, CEFR language levels (A1–C2) and EQF qualification levels.